

Miscellaneous

2018 - Why is the Indian Regional Navigational Satellite System (IRNSS) needed? How does it help in navigation? (10)

The **IRNSS**, operationally named **NavIC (Navigation with Indian Constellation)**, is India's indigenous satellite navigation system developed by Indian Space Research Organisation.

Need of IRNSS/NavIC

1. **Strategic Autonomy** - Dependence on foreign systems like **GPS (USA)**, **GLONASS (Russia)**, or **Galileo (EU)** poses security risks, as access can be denied during conflicts. Eg- During the **Kargil War (1999)**, USA denied GPS data to India.
2. **Sovereignty over Navigation** - Provides India independent and reliable **Position, Navigation, and Timing (PNT)** services over Indian territory and surrounding region.
3. **Regional Coverage** - Covers India and a region extending **1,500 km beyond its borders**, ensuring accurate navigation across South Asia and the **Indian Ocean Region**.
4. **Civilian Applications** - Terrestrial, aerial, and marine navigation; vehicle tracking; disaster management; mapping and geodetic surveys; mobile phone integration.
5. **Military Applications** - Missile guidance, troop movement, naval operations, border surveillance.
6. **Economic Benefits** - Supports precision agriculture, fisheries, transport logistics, and infrastructure development.

How NavIC Helps in Navigation

1. **Constellation** - **7 satellites (3 in Geostationary Orbit, 4 in Geosynchronous Orbit)** provide continuous coverage over the Indian region.
2. **Dual Frequency** - Operates on **L5 and S-band**, reducing errors caused by ionospheric delays and providing better accuracy than single-frequency systems.
3. **Accuracy** - Provides position accuracy better than **20 metres** in the primary service area and **10 metres** for restricted (military) service.
4. **Two Services** - **Standard Positioning Service (SPS)** for civilian use and **Restricted Service (RS)** for authorised users (military, strategic).
5. **Integration** with international systems like **GPS** for enhanced accuracy and reliability.
6. **Fishermen Safety** - Indian Space Research Organisation provides NavIC-based communication devices to fishermen for receiving emergency alerts and location-based services.

NavIC represents India's technological self-reliance in the strategic domain of satellite navigation, aligning with the vision of **Atmanirbhar Bharat**.

2025 - The ethos of civil service in India stand for the combination of professionalism with nationalistic consciousness - Elucidate. (10)

The Indian civil service, rooted in the **Weberian model of bureaucracy**, has evolved to combine **professional competence** with a deep commitment to **national development** and **constitutional values**.

Professionalism in Civil Service

1. **Merit-based recruitment** through the Union Public Service Commission ensures competence and impartiality. Eg- Competitive examination system for **IAS, IPS, IFS**.
2. Training institutions like Lal Bahadur Shastri National Academy of Administration (**Mussoorie**) instill administrative skills, ethics, and leadership. Eg- **'Bharat Darshan'** exposure tours.
3. Adherence to **rules, procedures**, and the principle of **political neutrality**.
4. Expertise in **policy formulation, implementation, and evaluation** across diverse sectors.
5. Commitment to **Rule of Law** - civil servants uphold constitutional provisions and legal frameworks.

Nationalistic Consciousness

1. The civil service was '**Indianised**' after independence, transforming from a colonial instrument of control to a vehicle of national development. Eg- Sardar Vallabhbhai Patel's vision of IAS as the '**steel frame**' of India.
2. **District administration model** - DMs serve as the face of governance in remotest areas, ensuring **national integration**.
3. Role in **nation-building** - Eg- land reforms, **Green Revolution**, poverty alleviation, literacy missions, **Aadhaar**, **Digital India**.
4. **Crisis management during national emergencies** - natural disasters, pandemics (**COVID-19 management** by district administration).
5. Constitutional values of **justice, liberty, equality, and fraternity** guide civil service conduct. Eg- Upholding **secularism** and protecting minority rights.
6. Commitment to **social justice** - implementation of reservation policies, tribal welfare, women's empowerment schemes.

Challenges

1. **Political interference** and transfers undermine neutrality.
2. **Red tape** and resistance to reform.
3. Ethical dilemmas in balancing political directives with public interest.

The Indian civil service, at its best, embodies what Jawaharlal Nehru envisioned: "**a service that is devoted to the nation, that understands the problems of the country, and that has the courage and integrity to serve the people.**"

**CIVILSDAILY IAS**

Smash Mains 2026

*A focused Mentorship to improve your GS Mains Score **from 300 to 450+***

TARGET: UPSC CSE MAINS 2026

Starts: 7 JUNE

1:1 Personalized Mentorship

Weekly mentor calls, tailored study plans & post-test course correction

12 GS FLTs + 10 Sectional Tests

70% Mains Hit Ratio, Model Answers & All India Rankings

X-Factor Classes + Notes

Smash Essay + Smash PYQ Enriched GS notes and strategy classes the topper-level edge

X-Factor Classes & Notes



1:1 Mentorship & Tests

For Smash Mains Enrollment



WhatsApp Over

80765 17832

Scan QR to know more about Smash Mains 2026

